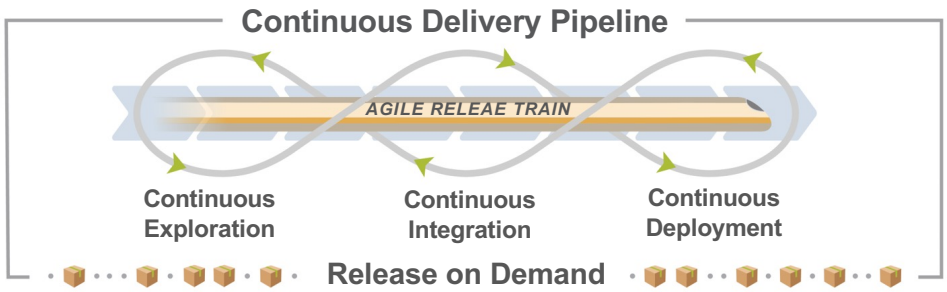
Building the Continuous Delivery Pipeline with DevOps

- The Continuous Delivery Pipeline (CDP) represents the workflows, activities, and automation needed to deliver new functionality more frequently.
- Each Agile Release Train builds and maintains, or shares, a pipeline.
- Organizations map their current pipeline into this new structure and remove delays and improve the efficiency of each step.



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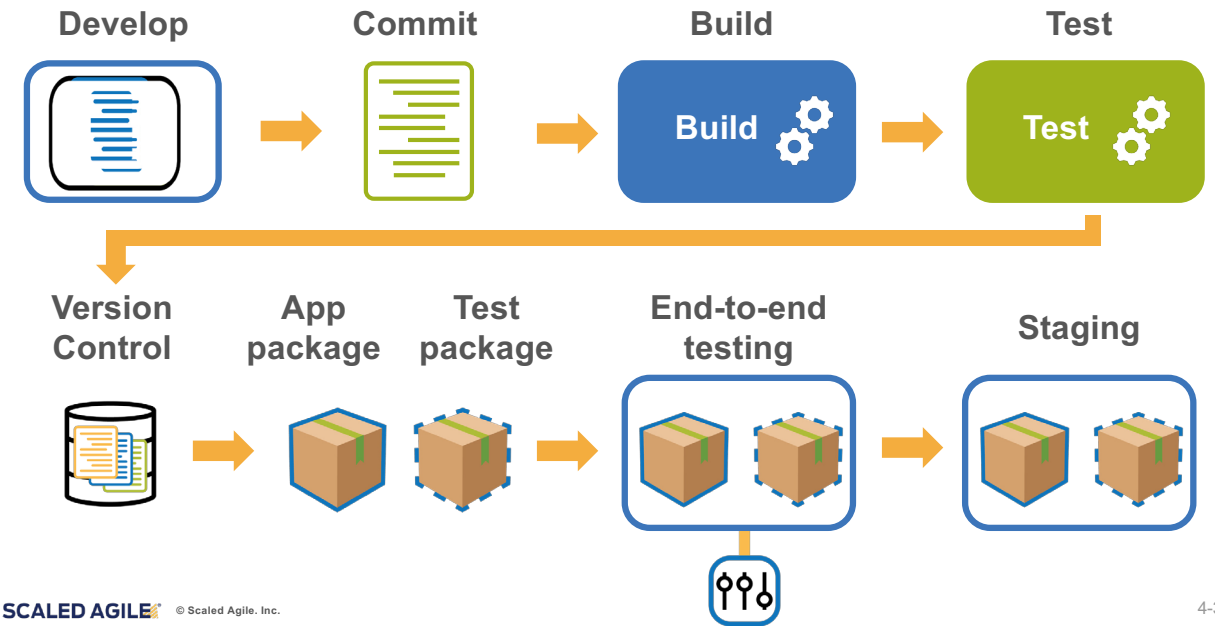
Continuous code integration



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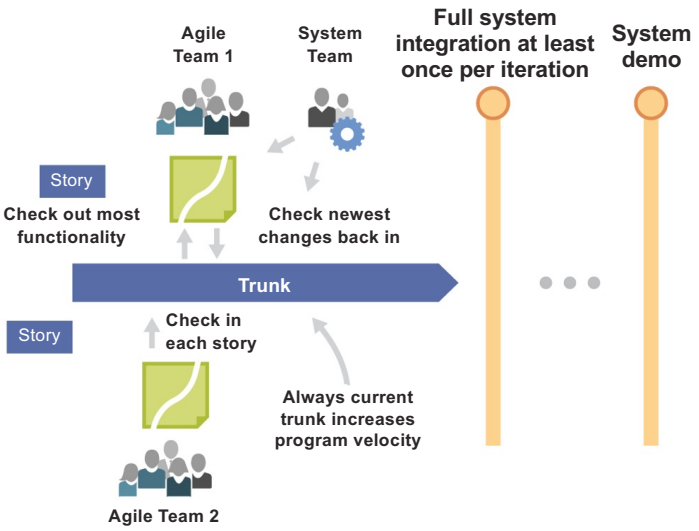
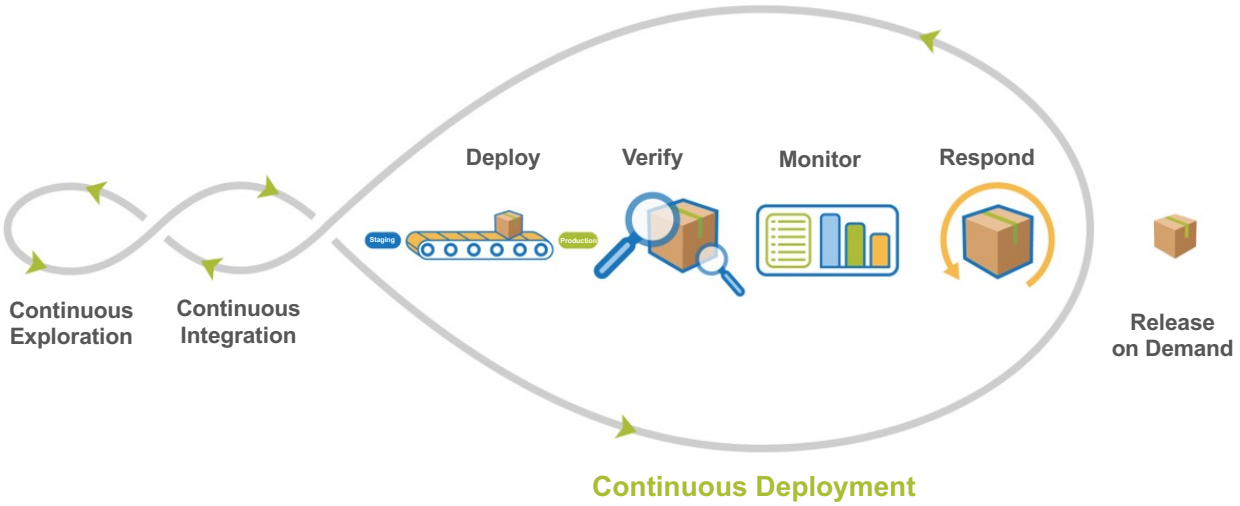
Trunk-based development

Teams continuously integrate assets (leaving as little as possible to the System Team).

- Avoid physical branching for software
- Frequently integrate hardware branches



Continuous Deployment – Getting to production early



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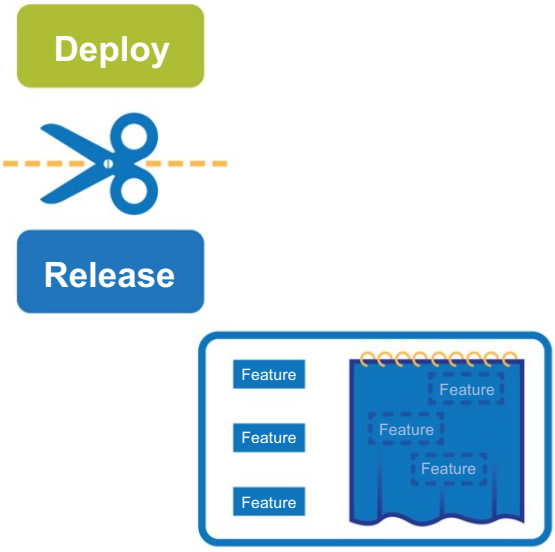
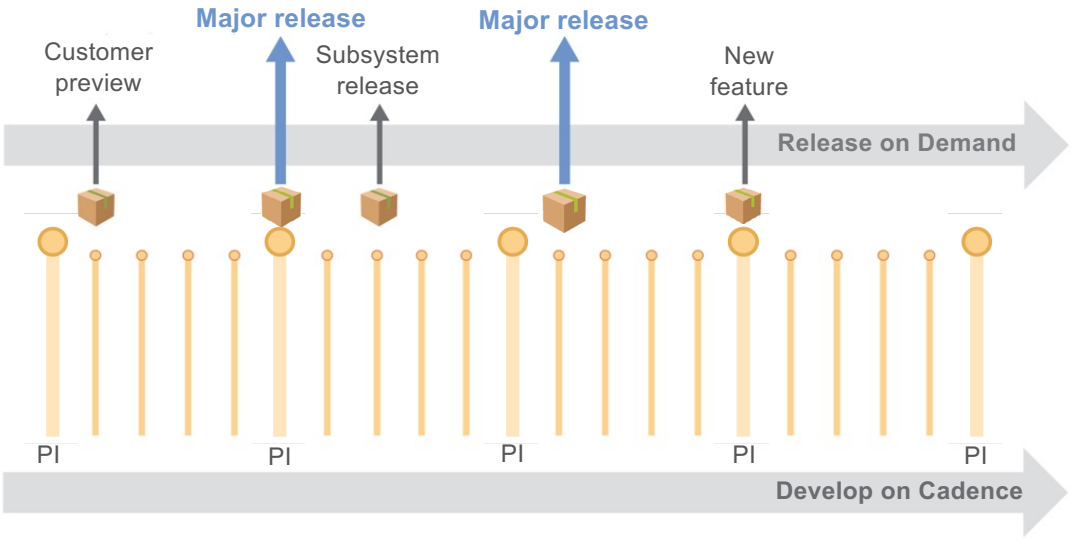
Separate deploy from release

- Separate deploy to production from release
- Hide new functionality under feature toggles
- Enables testing background and foreground processes in the actual production environment before exposing new functionality to users
- Timing of the release becomes a business decision



Develop on cadence. Release on Demand.



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Discussion: Continuous integration and deployment challenges

Prepare 15 min | Share 5 min

- **Step 1:** Working in your groups, think about the various aspects of environment, culture, tools, and people and discuss the following:
  - What are the challenges to continuously integrating?
  - What are the challenges to continuously deploying?
- **Step 2:** As a group, prepare a list of three to five items that make it hard to continuously integrate and deploy. What may be some ways to solve them?
- **Step 3:** Be prepared to share with the class.

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4.5 Improving flow with communication and synchronization

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Communication and synchronization with daily stand-ups

Basic daily stand-Up (DSU) agenda

Each person answers:

1. What did I do yesterday to advance the Iteration Goals?
2. What will I do today to advance the Iteration Goals?
3. Are there any impediments that will prevent the team from meeting the Iteration Goals?

The meet-after agenda

1. Review topics the Scrum Master wrote on the meet-after board
2. Involved parties discuss, uninvolved people leave

4-43

Activity: Reenact the daily stand-up (DSU)

You will participate in and observe a reenactment of the daily stand-up meeting. Four to five volunteers will play the role of *team member*. Your instructor will play the role of the *Scrum Master*.

- **Step 1**: As an observer, take notes and reflect on the following:
  - How long do you think the meeting should be?
  - Where should it take place?
  - What is the main purpose of the daily stand-up?
- **Step 2:** Share some of your insights as an observer

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4-44

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Reenact the Daily Stand-Up (DSU)

How long do you think the meeting should be?

Where should it take place?

What is the main purpose of Daily Stand-Up?

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Video: Backlog Refinement Workshop

Duration 5 min

https://bit.ly/Video-RunningBacklogRefinement

The backlog refinement workshop

The backlog refinement workshop is a preview and elaboration of upcoming Stories.

- Helps the team think about new Stories prior to Iteration Planning
- Provides enough time to identify and resolve dependencies and issues that could impact the next Iteration
- The team can improve Stories, add acceptance criteria, and identify missing information
- Most of the focus is on the next Iteration, but it allows time to discuss future Iterations and even Features for the next PI

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4.6 Demonstrating value with the Iteration Review

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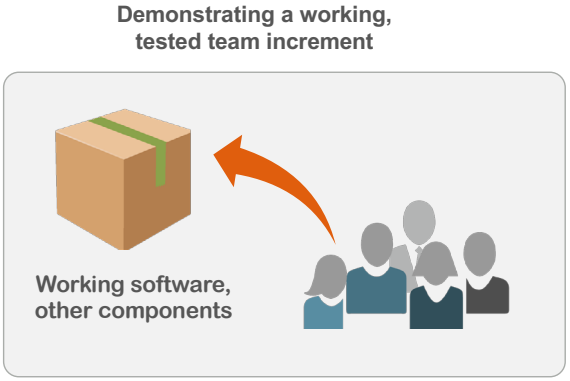
The Iteration Review

- The Iteration Review provides the true measure of progress by showing working software functionality, hardware components, etc.
- Preparation starts with planning
- Teams demonstrate every Story, spike*, refactor, and non-functional requirement (NFR)
- Attendees are the team and its stakeholders

*A spike is a research Story, considered an exploration style Enabler



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Iteration Review guidelines

- **Timebox:** 1 to 2 hours.
- **Preparation:** Review preparation should be limited to 1 to 2 hours. Minimize presentation. Work from the repository of Stories.
- **Attendees:** If a major stakeholder cannot attend, the Product Owner should follow up individually.

Sample Iteration Review Agenda

1. Review business context and Iteration Goals
2. Demo and solicit feedback of each Story, spike, refactor, and NFR
3. Discuss Stories not completed and why
4. Identify risks, impediments
5. Revise team backlog and Team PI Objectives as needed

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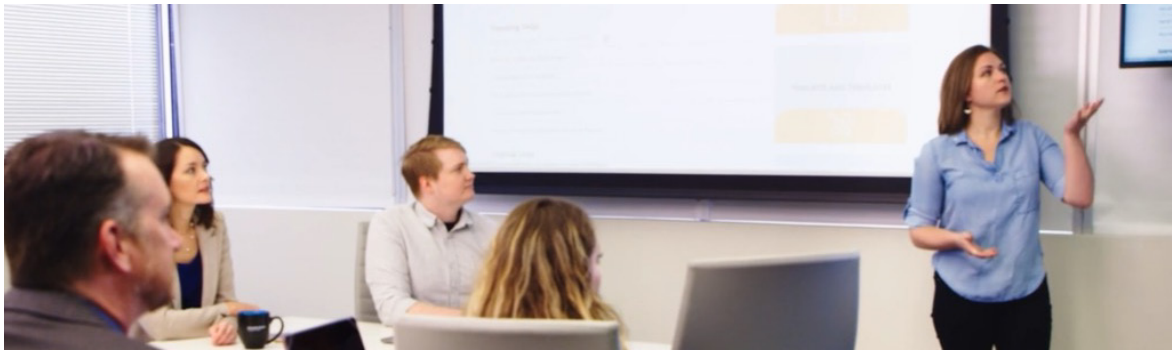
Two views from the Iteration Review based on a working system

- **How we did in the Iteration**
  - Did we meet the goal?
  - Story-by-Story review
- **How we are doing in the PI**
  - Review of PI Objectives
  - Review of remaining PI scope and reprioritizing if necessary



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Scalable definition of done

Team Increment	System Increment	Solution Increment	Release
• Stories satisfy acceptance criteria • Acceptance tests passed (automated where practical) • Unit and component tests coded, passed, and included in the BVT • Cumulative unit tests passed • Assets are under version control • Engineering standards followed • NFRs met • No must-fix defects • Stories accepted by Product Owner	• Stories completed by all teams in the ART and integrated • Completed features meet acceptance criteria • NFRs met • No must-fix defects • Verification and validation of key scenarios • Included in build definition and deployment process • Increment demonstrated, feedback achieved • Accepted by Product Management	• Capabilities completed by all trains and meet acceptance criteria • Deployed/installed in the staging environment • NFRs met • System end-to-end integration verification, and validation done • No must-fix defects • Included in build definition and deployment/transition process • Documentation updated • Solution demonstrated, feedback achieved • Accepted by Solution Management	• All capabilities done and meet acceptance criteria • End-to-end integration and solutions V&V done • Regression testing done • NFRs met • No must-fix defects • Release documentation complete • All standards met • Approved by Solution and Release Management

4-51

Discussion: What is your definition of done?

Prepare 5 min

Share 3 min

- **Step 1:** As a group, craft a definition of what it means to you to finish a Story
- **Step 2:** Considering the criteria in the Team Increment described in the previous slide, discuss some criteria that comprise your definition of done
- **Step 3:** Be prepared to share with the class

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4.7 Improving with the Iteration Retrospective

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Iteration Retrospective

- **Timebox:** an hour or less
- **Purpose:** Pick one or two items that can be improved upon for the next Iteration
- **Outcome:** Enter the improvement items in the Team Backlog

Sample Agenda

Part 1: Quantitative

1. Review the improvement backlog items targeted for this Iteration. Were they all accomplished?
2. Did the team meet the goals (yes/no)?
3. Collect and review the agreed-to Iteration print metrics.

Part 2: Qualitative

1. What went well?
2. What didn't?
3. What can we do better next time?

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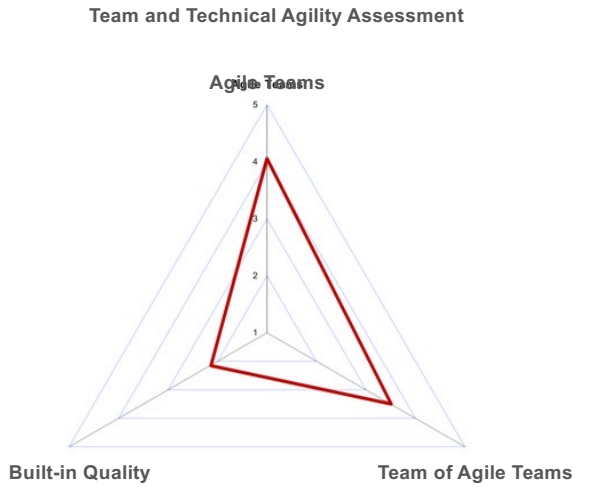
Team and Technical Agility (TTA) competency assessment

- Periodically measure the progress being made toward the three TTA dimensions
- Identify specific practices for potential improvement
- Reassess periodically to observe trends



Assessments are available on the SAFE Community Platform:
https://bit.ly/Community-MeasureAndGrow

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Sample Iteration Metrics

Functionality	Iteration 1	Iteration 2	Iteration 3
Velocity planned			
Velocity actual			
# Stories planned			
# Stories accepted			
% Stories accepted			
Quality			
Unit test coverage %			
# Defects			
# New test cases			
# New test cases automated			
Total tests			
Total % tests automated			
# Refactors			

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Activity: Simulate an Iteration Retrospective

Prepare: 8 min

Share: 2 min

In your group, facilitate an Iteration retrospective of this course so far.

- **Step 1**: Pick someone in your group to play the role of the Scrum Master to facilitate the Iteration retrospective event
- **Step 2:** As a group, participate in the retrospective by discussing the following:
  - What went well?
  - What didn't go so well?
  - What can be done better?
- **Step 3:** Share some of your group's insights with the class

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Action Plan: Executing the Iteration

Prepare: 5 min

Share: 3 min

- **Step 1:** As a team, brainstorm one to three actions you could take to improve in any areas related to this lesson
- **Step 2:** Individually write down at least one improvement item
- **Step 3:** Share one item you discussed as a team and one item you individually wrote in your Action Plan

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Executing the Iteration

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Lesson review

In this lesson you:

- Visualized your current flow of work
- Learned how to apply burn-up and cumulative flow diagrams for measuring flow
- Reflected on the different aspects of Built-in Quality
- Recognized the importance of a continuous delivery pipeline
- Experienced the communication and synchronization benefits of the daily stand-up
- Explained how to run a successful backlog refinement session
- Discussed the purpose of the Iteration Review and created a definition of done
- Participated in a retrospective and identified improvements

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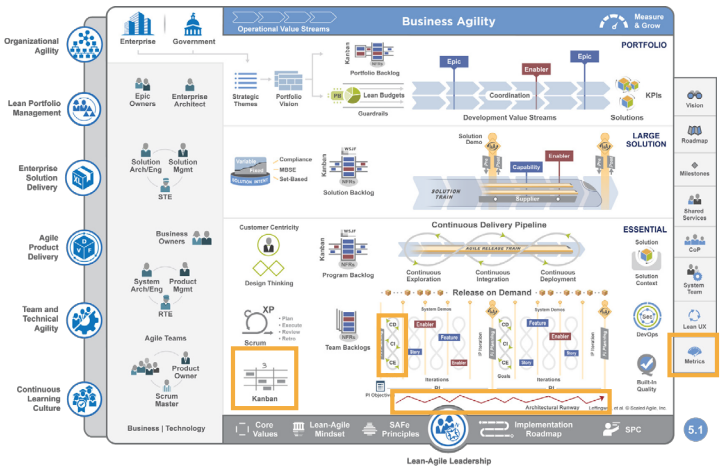
Articles used in this lesson

Read these Framework articles to learn more about topics covered in this lesson

- "Kanban"
  https://www.scaledagileframework.com/team-kanban/
- "Metrics"
  https://www.scaledagileframework.com/metrics/
- "Architectural Runway"
  https://www.scaledagileframework.com/architectural-runway/
- "Continuous Integration"
  https://www.scaledagileframework.com/continuous-integration/



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Continue your SAFe journey with the following resources:

Watch this three-part video series, *Kanban for Teams*, to walk through the processes of setting up, implementing, and incrementally improving a team Kanban. https://bit.ly/Playlist-KanbanTeams	Use the SAFe Collaborate template, "Determine the Team's Definition of Done," with your team to agree upon guardrails, rules, and processes to ensure built-in quality for your solutions. https://bit.ly/Template-DetermineDoD
Use the SAFe Collaborate template "Experience Batch Sizes" with your team to explore the impact of reducing batch sizes on overall throughput. https://bit.ly/Template-BatchSize	Watch this sixty-minute video, *Community Webinar: DevSecOps in Real Life,* to delve into the most important practices that fuel the Continuous Delivery Pipeline and how they are implemented in real life. https://bit.ly/Community-DevSecOpsWebinar
Watch this five-part video series, *Running Effective Iteration Events,* to explore each of the key team-level events in SAFe. https://bit.ly/Playlist-EffectiveIterationEvents	Use the SAFe Collaborate template, "Retrospective Plus, Minus, Delta," to run an effective retrospective with your team. https://bit.ly/Template-PlusMinusDelta

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Lesson notes

Enter your notes below. If using a digital workbook, save your PDF often so you don't lose any of your notes.

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Lesson 5

Executing the Program Increment (PI)

SAFe® Course - Attending this course gives learners access to the SAFe® Practitioner exam and related preparation materials.

SP SAFe® 5

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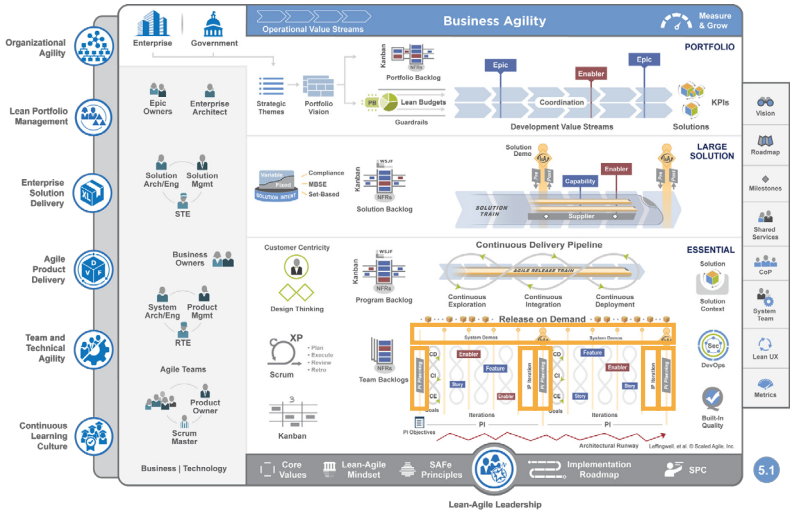
Lesson Topics

5.1 PI Planning

5.2 ART Sync and System Demo

5.3 Innovation and Planning (IP) Iteration and the Inspect & Adapt event



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Learning objectives

At the end of this lesson, you should be able to:

- Explain the benefits of planning together as a whole ART
- Participate in a PI Planning event
- Recognize the importance of integrating and demonstrating together with the System Demo
- Recognize the need for the IP Iteration to provide time and space for the ART to learn together
- Explain how to improve as an ART through the Inspect and Adapt event

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5.1 PI Planning

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Video: Introduction to PI Planning

Duration 3 min

https://bit.ly/Video-PIPlanningOverview

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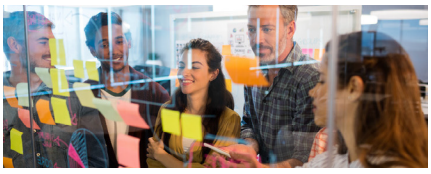
5-5

What is PI Planning?

PI Planning is a cadence-based event that serves as the heartbeat of the Agile Release Train (ART), aligning all teams on the ART to a shared mission and Vision.

- Two days every 8 – 12 weeks (10 weeks is typical)
- Everyone plans together
- Product Management owns Feature priorities
- Development teams own Story planning and high-level estimates
- Architect/Engineering and UX work as intermediaries for governance, interfaces, and dependencies

PI Planning



Agile Team

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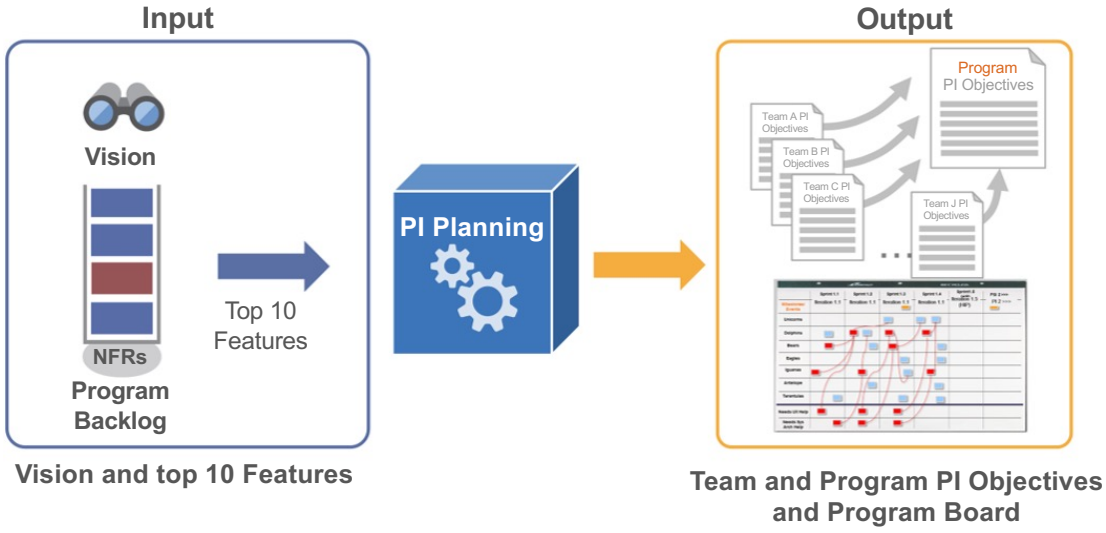
The PI Planning process



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Create Alignment with PI Objectives

- Objectives are business summaries of what each team intends to deliver in the upcoming PI.
- They often directly relate to intended Features in the backlog.
- Other examples:
  - Aggregation of a set of Features
  - A Milestone like a trade show
  - An Enabler Feature supporting the implementation
  - A major refactoring

Objectives for PI 1	BV	AV
1. Show routing calculations between the 5 most frequent destinations		
2. Navigate autonomously from distribution center to the most frequent destination		
3. Parallel park for a delivery		
4. Return to the distribution center after delivery		
5. Include traffic data in route planning		
6. Recall a delivery that is already in progress		
Uncommitted Objectives		
7. Spike: Reduce GPS signal loss by 25%		
8. Demonstrate real-time rerouting to avoid delays (e.g., accident, construction)		

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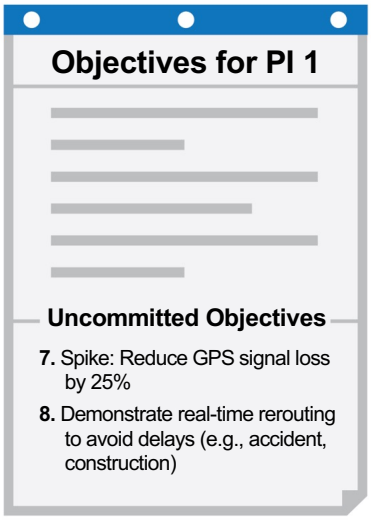
Maintain predictability with uncommitted objectives

Uncommitted objectives help improve the predictability of delivering business value.

- They are planned and aren't extra things teams do 'just in case you have time'
- They are not included in the commitment, thereby making the commitment more reliable
- If a team has low confidence in meeting a PI Objective, it should be moved to uncommitted
- If an objective has many unknowns, consider moving it to uncommitted and put in early spikes
- Uncommitted objectives count when calculating load



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SMART team PI Objectives

- **S**pecific - States the intended outcome as simply, concisely, and explicitly as possible (Hint: Try starting with an action verb).
- **M**easurable - It should be clear what a team needs to do to achieve the objective. The measures may be descriptive, yes/no, quantitative, or provide a range.
- **A**chievable - Achieving the objective should be within the team's control and influence
- **R**ealistic - Recognize factors that cannot be controlled.(Hint: Avoid making overly optimistic assumptions)
- **T**ime-bound - The time period for achievement must be within the PI, and, therefore, all objectives must be scoped appropriately.

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Activity: Identify ART roles

Duration: 3 min

- **Step 1:** Identify ART roles for the simulation
- **Step 2:** Ensure that you have all key roles required for the PI Planning simulation

Simulation role	Assigned to
Executive	Volunteer
Product Manager	Volunteer
System Architect, UX, Development Manager	Volunteer

Note: Your Instructor will play the role of the RTE.

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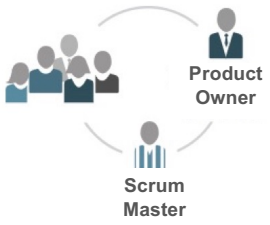
Activity: Identify team names and roles

Duration: 3 min

- **Step 1:** Your team is your group. Create a team name.
- **Step 2:** Select a Scrum Master for your team.
- **Step 3:** Select a Product Owner for your team.
- **Step 4:** Make sure the team's name and the names of the people selected are visible to all other teams.



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Simulation: Why are we here?

Alignment to a common mission

We are here to gain alignment and commitment around a clear set of prioritized objectives. I will now review the agenda for the next two days of the PI Planning event.

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5-13

Simulation: Day 1 agenda

Business context	8:00 – 9:00	• State of the business
Product/Solution Vision	9:00 – 10:30	• Vision and prioritized Features
Architecture Vision and development practices	10:30 – 11:30	• Architecture, common frameworks, etc. • Agile tooling, engineering practices, etc.
Planning context and lunch	11:30 – 1:00	• Facilitator explains the planning process
Team breakouts	1:00 – 4:00	• Teams develop draft plans and identify risks and impediments • Architects and Product Managers circulate
Draft plan review	4:00 – 5:00	• Teams present draft plans, risks, and impediments
Management review and problem solving	5:00 – 6:00	• Adjustments made based on challenges, risks, and impediments

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Simulation: Day 2 agenda

Planning adjustments	8:00 – 9:00	• Planning adjustments made based on previous day's management meeting
Team breakouts	9:00 – 11:00	• Teams develop final plans and refine risks and impediments • Business Owners circulate and assign business value to team objectives
Final plan review and lunch	11:00 – 1:00	• Teams present final plans, risks, and impediments
Program risks	1:00 – 2:00	• Remaining program-level risks are discussed and ROAMed
PI confidence vote	2:00 – 2:15	• Team and program confidence vote
Plan rework if necessary	2:15 – ???	• If necessary, planning continues until commitment is achieved
Planning retrospective and moving forward	After commitment	• Retrospective • Moving forward • Final instructions

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Simulation: Briefings

Executive

Product Manager

System Architect

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Simulation: Planning guidance

Expect this first PI Planning to feel a bit chaotic. Future PI Planning meetings will become more routine.

Product Owners: You have the content authority to make decisions at the user Story level

Scrum Masters: Your responsibility is to manage the timebox, the dependencies, and the ambiguities

Agile Team: Your responsibility is to define user Stories, plan them into the Iteration, and work out interdependencies with other teams

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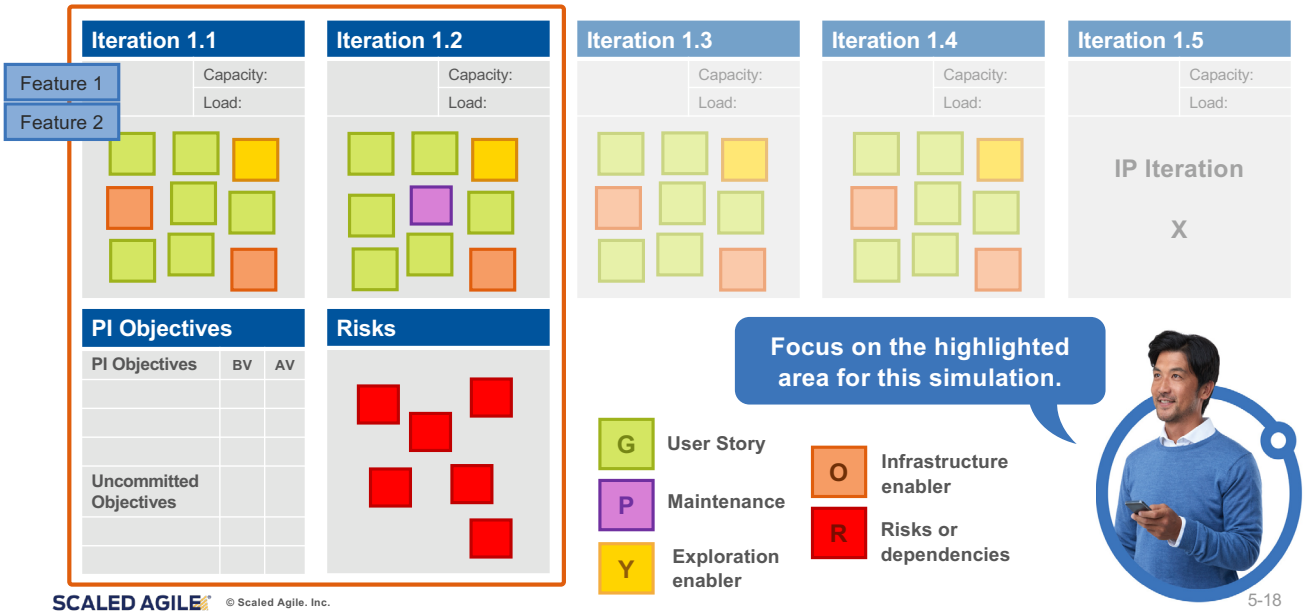
5-17

Simulation: Planning requirements



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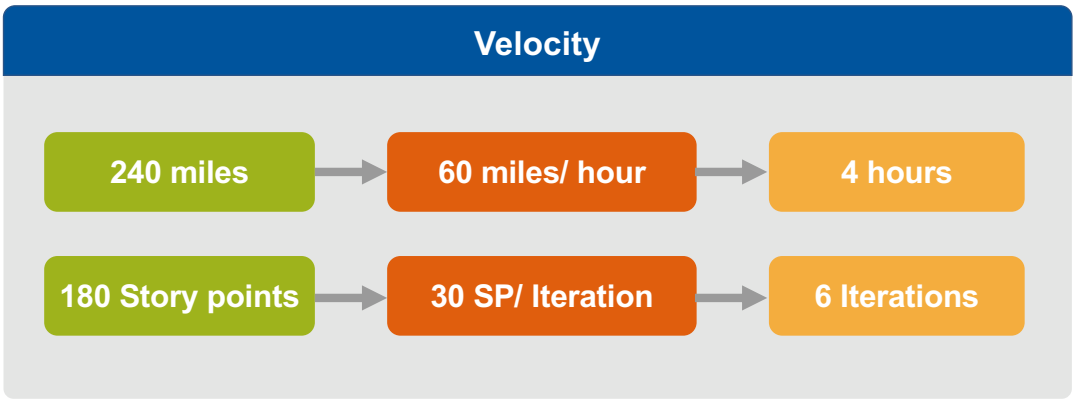
Simulation: Using historical data to calculate velocity



Establish velocity by looking at the average output of the last Iterations.

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Simulation: Calculate your capacity

Calculating Iteration capacity

- For every full-time Agile Team member contributing to Solution development, give the team 8 points (adjust for part-timers).
- Subtract 1 point for every team member vacation day and holiday.
- Find a small Story that would take about a half day to develop and a half day to test and validate. Call it a 1.
- Estimate every other Story relative to that one.

Example:
A 7-person team composed of 3 developers, 2 testers, 1 Product Owner, and 1 Scrum Master
Exclude the Scrum Master, Product Owner, and vacation time from the calculation
Calculated capacity: 5 x 8 points = 40 points per Iteration

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Activity: Calculate your capacity

Duration: 5 min

- **Step 1:** Review the example on the previous slide
- **Step 2:** Calculate your own capacity for the next two, 2-week Iterations
  - The first Iteration starts Monday
  - Use your real availability
- **Step 3:** Make sure you have your group's capacity calculated

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Activity: Team breakout #1

Duration: 50 min

You will be planning a short Program Increment with two Iterations.

- **Step 1:** Setup the workspace. Enter the capacity for each Iteration.
- **Step 2:** Pick up a Feature from the Product Manager.
- **Step 3:** Estimate the Stories using Story points.
- **Step 4:** Load the Stories into the Iterations.
- **Step 5:** Write the PI Objectives using clear statements.
- **Step 6:** Identify the uncommitted objectives.
- **Step 7:** Identify any program risks and dependencies.

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Activity: Scrum of Scrums (SoS)

Duration: 5 min

- **Step 1:** Observe the SoS, conducted by the RTE
- **Step 2:** Each Scrum Master provides the current status and addresses the questions from the RTE
- **Step 3:** The RTE holds a meet-after after the sync (limited to 1 – 2 topics for the simulation)

Scrum of Scrums questions are on the following slide.

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Activity: Scrum of Scrums (SoS)

Duration: 5 min

SoS Sync Questions	Team 1	Team 2	Team 3	Team 4	Team 5
Have you identified the capacity for each Iteration of the PI?					
Have you identified most of the Stories for the first two Iterations and begun estimating?					
Have you begun resolving dependencies with other teams?					
Are you discussing tradeoffs and conflicting priorities with your Business Owners?					
Have you identified any program risks?					
Will you be ready to start writing PI Objectives in the next 15 minutes?					
Is there anything you need to discuss with other Scrum Masters? If so, stay for the meet-after.					

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Activity: Draft plan review

Duration: 10 min

- **Step 1:** Present the summary of your team's first two Iterations and one or more draft PI Objectives
- **Step 2:** Make sure that you have included the following:
  - Capacity and load for each Iteration
  - Draft PI Objectives
  - Program risks and impediments

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Management review and problem-solving

At the end of day 1, management meets to make adjustments to scope and objectives based on the day's planning.

Common questions during the managers' review:

- What did we just learn?
- Where do we need to adjust? Vision? Scope? Team assignments?
- Where are the bottlenecks?
- What Features must be de-scoped?
- What decisions must we make between now and tomorrow to address these issues?

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Activities during Day 2

Day 1	
Business context	8:00–9:00
Product/Solution Vision	9:00–10:30
Architecture Vision and development practices	10:30–11:30
Planning context and lunch	11:30–1:00
Team breakouts	1:00–4:00
Draft plan review	4:00–5:00
Management review and problem solving	5:00–6:00

Day 2	
Planning adjustments	8:00–9:00
Team breakouts	9:00–11:00
Final plan review and lunch	11:00 –1:00
Program risks	1:00–2:00
PI confidence vote	2:00–2:15
Plan rework if necessary	2:15–???
Planning retrospective and moving forward	After commitment

5-27

Make planning adjustments

- Based on the previous day's management review and problem-solving meeting, adjustments are discussed.
- Possible changes:
  - Business priorities
  - Adjustment to Vision
  - Changes to scope
  - Realignment of work and teams

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Team breakout #2

Based on new knowledge and a good night's sleep, teams work to create their final plans.

- In the second team breakout, Business Owners circulate and assign business value to PI Objectives from low (1) to high (10)
- Teams finalize the Program Increment plan
- Teams also consolidate program risks, impediments, and dependencies
- Uncommitted objectives provide the capacity and guard band needed to increase the reliability of cadence-based delivery

Objectives for PI 1	BV	AV
1. Show routing calculations between the 5 most frequent destinations	10	
2. Navigate autonomously from distribution center to the most frequent destination	8	
3. Parallel park for a delivery	7	
4. Return to the distribution center after delivery	10	
5. Include traffic data in route planning	7	
6. Recall a delivery that is already in progress	7	
Uncommitted Objectives		
7. Spike: Reduce GPS signal loss by 25%	2	
8. Demonstrate real-time rerouting to avoid delays (e.g., accident, construction)	5	

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Video: Program Board

Duration: 7 min

https://bit.ly/Video-ProgramBoard

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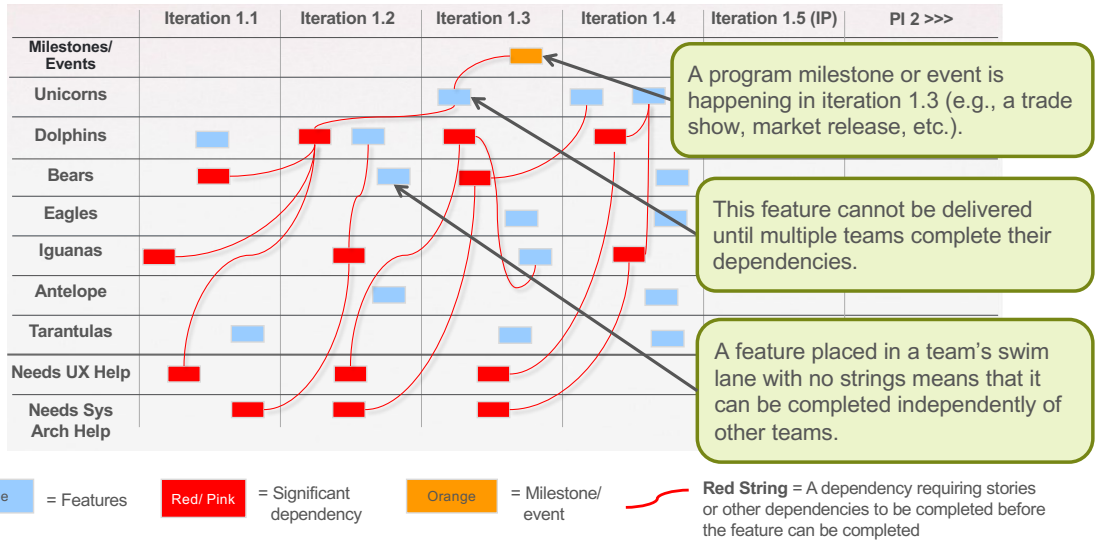
Program board: Feature delivery, dependencies, and Milestones



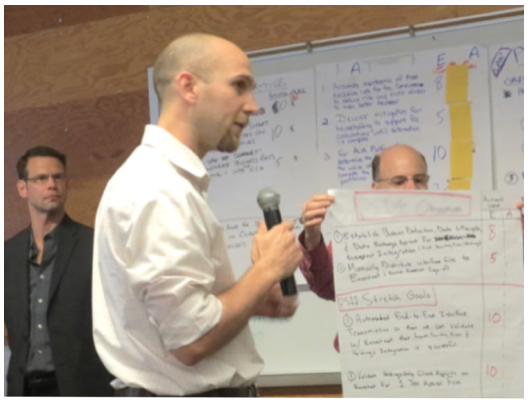
5-31

Final plan review

Teams and Business Owners peer-review all final plans.

Final plan review agenda

1. Changes to capacity and load
2. Final PI Objectives with business value
3. Program risks and impediments
4. Q&A session



Final plan review

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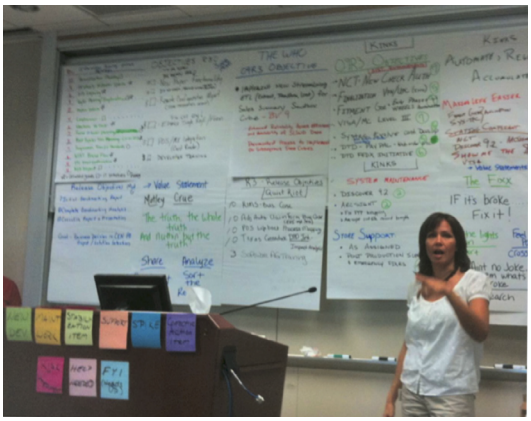
Building the final plan

- Final plans are reviewed by all teams
- Business Owners are asked whether they accept the plan
- If so, the plan is accepted
- If not, the plans stay in place, and the team continues planning after the review



A team presenting their final plan

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5-33

Addressing program risks

After all plans have been presented, remaining program risks and impediments are discussed and categorized.

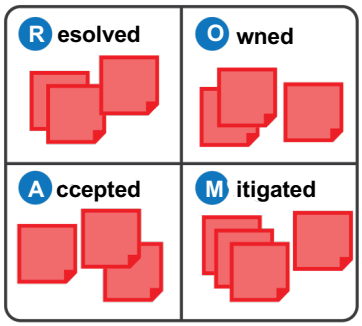
ROAMing risks:

Resolved - Has been addressed. No longer a concern.

Owned - Someone has taken responsibility.

Accepted - Nothing more can be done. If risk occurs, release may be compromised.

Mitigated - Team has plan to adjust as necessary.



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Confidence vote: Team and program

After dependencies are resolved and risks are addressed, a confidence vote is taken by the team and program.

A commitment with two parts:

1. Teams agree to do everything in their power to meet the agreed-to objectives
2. In the event that fact patterns dictate that it is simply not achievable, teams agree to escalate immediately so that corrective action can be taken

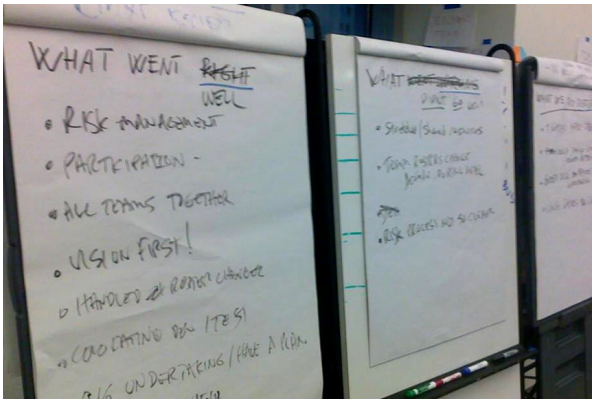
No confidence	Little confidence	Good confidence	High confidence	Very high confidence

Run a planning meeting retrospective

The PI planning event will evolve over time. Ending with a retrospective will help continuously improve it.

The PI Planning Event Retrospective

1. What went well
2. What didn't
3. What we can do better next time



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5.2 ART Sync and System Demo

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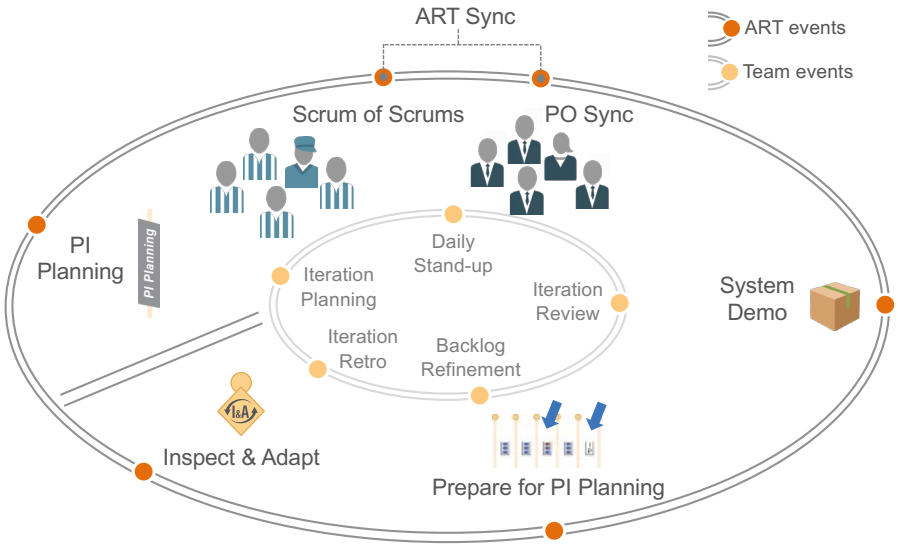
ART events drive the train

ART events create a closed-loop system to keep the train on the tracks.



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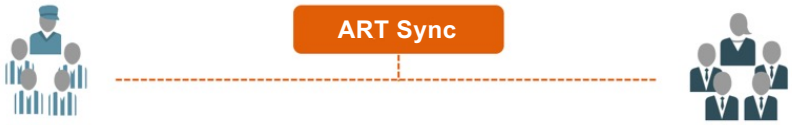
ART Sync is used to coordinate progress



Scrum of Scrums

- Visibility into progress and impediments
- Facilitated by RTE
- Participants: Scrum Masters, other select team members, SMEs if necessary
- Weekly or more frequently, 30-60 minutes
- Timeboxed and followed by a 'Meet After'

PO Sync

- Visibility into progress, scope, and priority adjustments
- Facilitated by RTE or PM
- Participants: PMs, POs, other stakeholders, and SMEs as necessary
- Weekly or more frequently, 30-60 minutes
- Timeboxed and followed by a 'Meet After'

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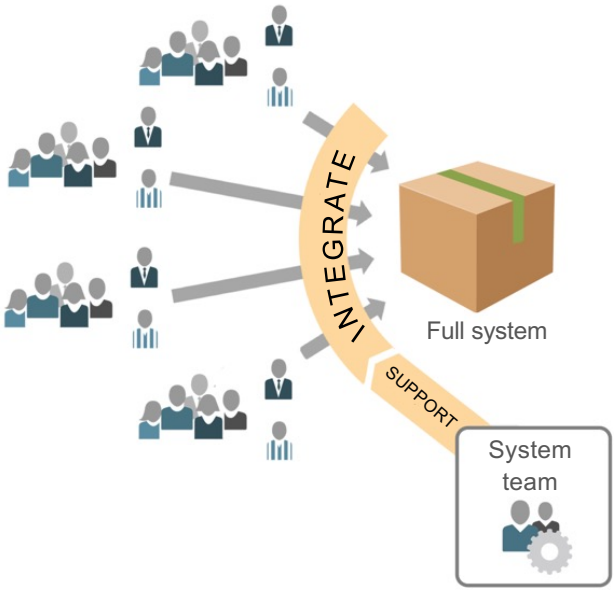
5-39

Demo the full system increment every two weeks

- Happens after the Iteration review (may lag by as much as one Iteration, maximum)
- Features are functionally complete or 'toggled' so as not to disrupt demonstrable functionality
- New Features work together and with existing functionality
- Demo from a staging environment which resembles production as much as possible



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Discussion: System Demo challenges

Prepare 3 min

Share 2 min

- **Step 1:** As a team, discuss the following:
  - What are challenges to having a new system increment every two weeks?
- **Step 2:** List three to five challenges and some ways to solve them
- **Step 3:** Be prepared to share with the class

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5.3 Innovation and Planning (IP) Iteration and the Inspect & Adapt event

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Innovation and Planning (IP) Iteration

Provide sufficient capacity margin to enable cadence. —Donald G. Reinertsen

Facilitate reliability, Program Increment readiness, planning, and innovation

- **Innovation:** Opportunity for innovation, hackathons, and infrastructure improvements
- **Planning:** Provides for cadence-based planning
- Estimating **guard band** for cadence-based delivery

> "Provide sufficient capacity margin to enable cadence."
>
> —Donald G. Reinertsen

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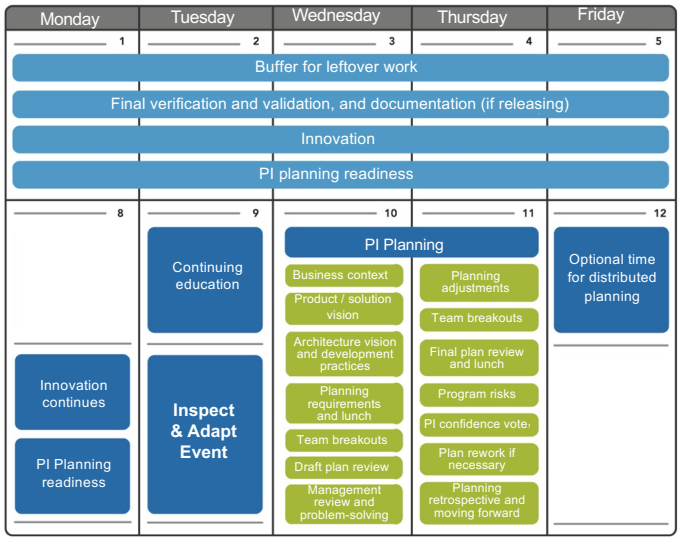
Example IP Iteration calendar



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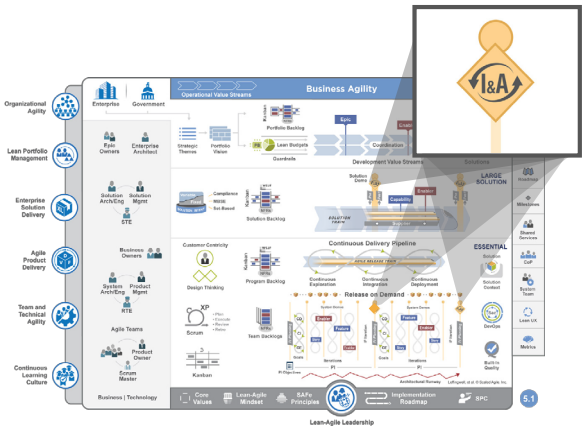
Improving results with the Inspect and Adapt event

Three parts of Inspect and Adapt:

1. **The PI System Demo**
2. **Quantitative and Qualitative Measurement**
3. **Problem-Solving Workshop**

- **Timebox:** 3 – 4 hours per PI
- **Attendees:** Teams and stakeholders



1. PI System Demo

At the end of the PI, teams demonstrate the current state of the Solution to the appropriate stakeholders.

- Often led by Product Management, POs, and the System Team
- Attended by Business Owners, ART stakeholders, Product Management, RTE, Scrum Masters, and teams

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Program performance reporting

Prior to or as part of the PI System Demo, teams review the business value achieved for each of their PI Objectives.

- Teams meet with their Business Owners to self-assess the business value they achieved for each objective
- Each team's planned vs actual business value is then rolled up to the program predictability measure.

Objectives for PI 3	Business Value: Plan	Business Value: Actual
Structured locations and validation of locations	7	7
Build and demonstrate a proof of concept for context images	8	8
Implement negative triangulation by: tags, companies and people	8	6
Speed up indexing by 50%	10	5
Index 1.2 billion more web pages	10	8
Extract and build URL abstracts	7	7
Uncommitted Objectives		
Fuzzy search by full name	7	0
Improve tag quality to 80% relevance	4	4
Totals	50	45
% Achievement:	**90%**	

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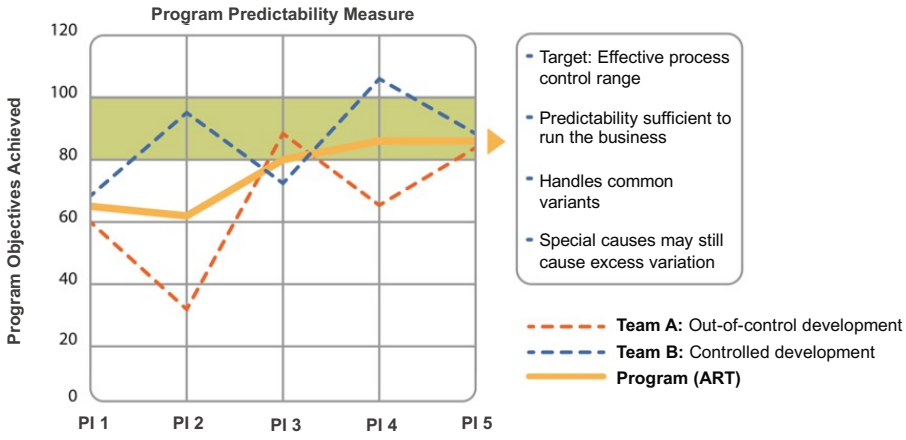
Measure ART Predictability

The Program Predictability measure compares actual business value achieved to planned business value.

Flow



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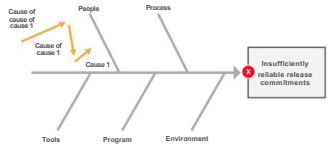
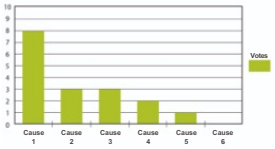
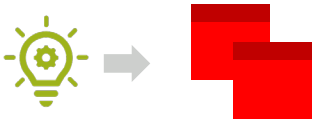
3. The Problem-Solving Workshop

Agree on the problem to solve	Apply root-cause analysis and 5 whys	Identify the biggest root-cause using Pareto analysis
X Insufficiently reliable release commitments		
Restate the new problem for the biggest root-cause	**Brainstorm solutions**	**Identify improvement backlog items**
X Insufficient architectural runway		

Action Plan: Executing the PI

Prepare 5 min Share 3 min

- **Step 1:** As a team, brainstorm one to three actions you could take to improve in any areas related to this lesson
- **Step 2:** Individually write down at least one improvement item
- **Step 3:** Share one item you discussed as a team and one item you individually wrote in your Action Plan

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Executing the PI

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Lesson review

In this lesson you:

- Explored the benefits of planning together as a whole ART
- Participated in a PI Planning event
- Recognized the importance of integrating and demonstrating together with the System Demo
- Recognized the need for the IP Iteration to provide time and space for the ART to learn together
- Explored how to improve as an ART through the Inspect and Adapt event

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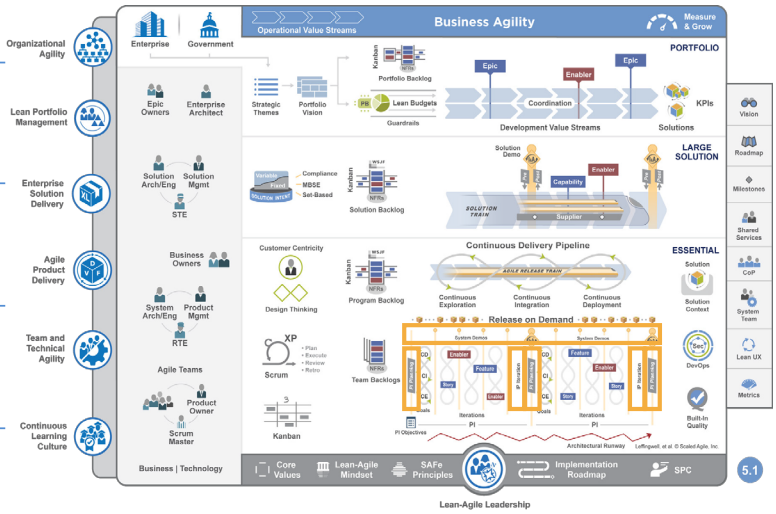
Articles used in this lesson

Read these Framework articles to learn more about topics covered in this lesson

- "PI Objectives" https://www.scaledagileframework.com/pi-objectives/
- "Innovation and Planning Iteration" https://www.scaledagileframework.com/innovation-and-planning-iteration/
- "Inspect and Adapt" https://www.scaledagileframework.com/inspect-and-adapt/
- "System Demo" https://www.scaledagileframework.com/system-demo/



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Continue your SAFe journey with the following resources:

Watch this quick one-minute video, *SAFe Developer Stories: My First PI Planning*, to hear about Elizabeth Flournoy's first PI Planning experience. **https://bit.ly/Video-FirstPIPlanning**	Download the "SAFe PI Planning Toolkit 5.1" for resources to support preparation, coordination, and communication to guide an ART through its PI Planning event. **https://bit.ly/Community-ToolkitsandTemplates**
Watch this six-minute video, *Preparation for PI Planning*, to learn about three areas an ART should focus on when preparing for PI Planning. **https://bit.ly/Video-PIPlanningPrep**	Download and use the resources found in the "SAFe PI Execution Toolkit" to support an ART in successful PI Execution. **https://bit.ly/Community-ToolkitsandTemplates**
Watch and share the videos in this playlist, "Agile Software Engineering Vlog" to learn how to move from software engineering to Agile software engineering. **https://bit.ly/Playlist-AgileSoftwareEngineering**	

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Lesson notes

Enter your notes below. If using a digital workbook, save your PDF often so you don't lose any of your notes.

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Lesson 6

Practicing SAFe

SAFe® Course - Attending this course gives students access to the SAFe® Practitioner exam and related preparation materials.

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Video: SAFe Certification Benefits

Duration 3 min

https://bit.ly/Video-SAFeCertificationBenefits

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A Path Towards Certification

Access exam study guides and practice tests

Download your certificate of course completion

Take the **Certification Exam**

Showcase your **Digital Badge** and get recognized as Certified SAFe Professional

Becoming a Certified SAFe Professional

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6-3

Video: Welcome to the SAFe Community Platform

Duration 3 min

https://bit.ly/Video-WelcomeSAFeCommunityPlatform

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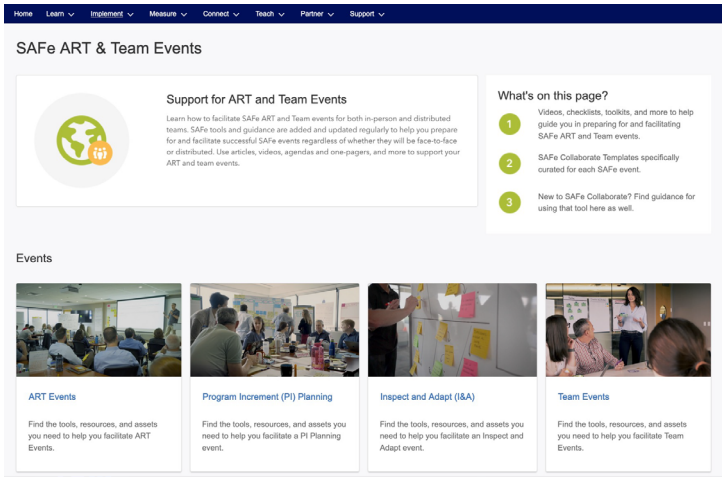
SAFe ART and Team Events

SAFe ART and Team Events: Use checklists, guides, templates, videos, and more to support your ART and Team Events



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SAFe Toolkits

Explore ready-to-use templates and job resources to help execute SAFe events and workshops more effectively

- SAFe® Remote ARTs Toolkit
- SAFe® Measure and Grow Workshop Toolkit
- SAFe® Iteration Execution Toolkit
- SAFe® PI Planning Toolkit
- SAFe® PI Execution Toolkit
- Team Formation Toolkit

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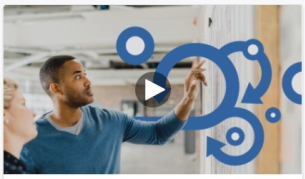
E-learning Resources

Discover and develop skills through self-paced, interactive e-learning modules to achieve your personal and professional goals

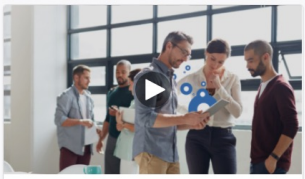


Agile Basics

E-learning

Learn what Agile is, where it comes from, why it continues to be used and needed, and how it supports teams and organizations to do what they do better.

30 - 45 Minutes

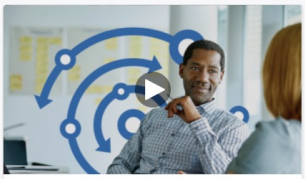


What is SAFe for Lean Enterprises

E-learning

Become more familiar with the goals and methods of SAFe to achieve Business Agility.

15 - 30 Minutes



SAFe Foundations: Core Values

E-learning

Build your understanding of the core values of SAFe and how they are applied in practice.

15 - 30 Minutes

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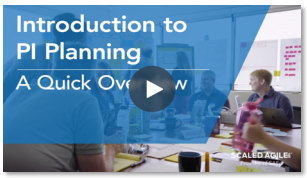
6-7

Community Video Hub

Access videos to support your learning and grow your skills



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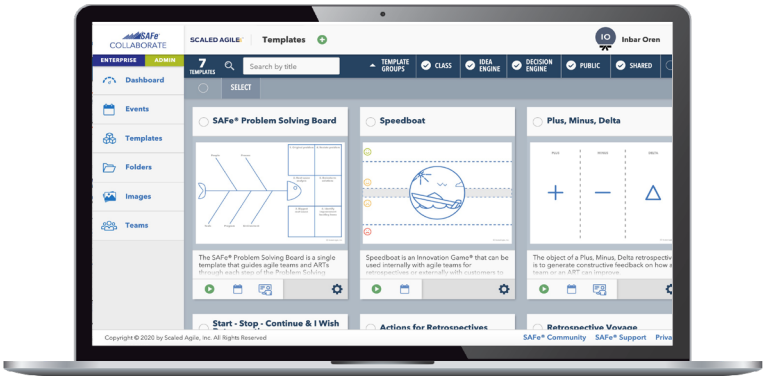
SAFe Collaborate

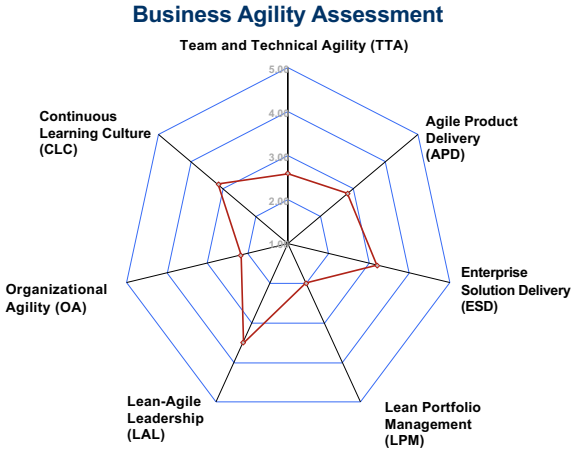
Organize and run virtual SAFe events in real-time

SAFe Collaborate is a visual, cloud-based workspace where organizations can orchestrate virtual SAFe events activities easily and effectively with predesigned and customizable templates.



Measure and Grow

Evaluate progress towards business agility with the SAFe assessments, Measure and Grow workshop and our assessment partners



agilityhealth® enabling business agility

comparative agility

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Team and Technical Agility Assessment

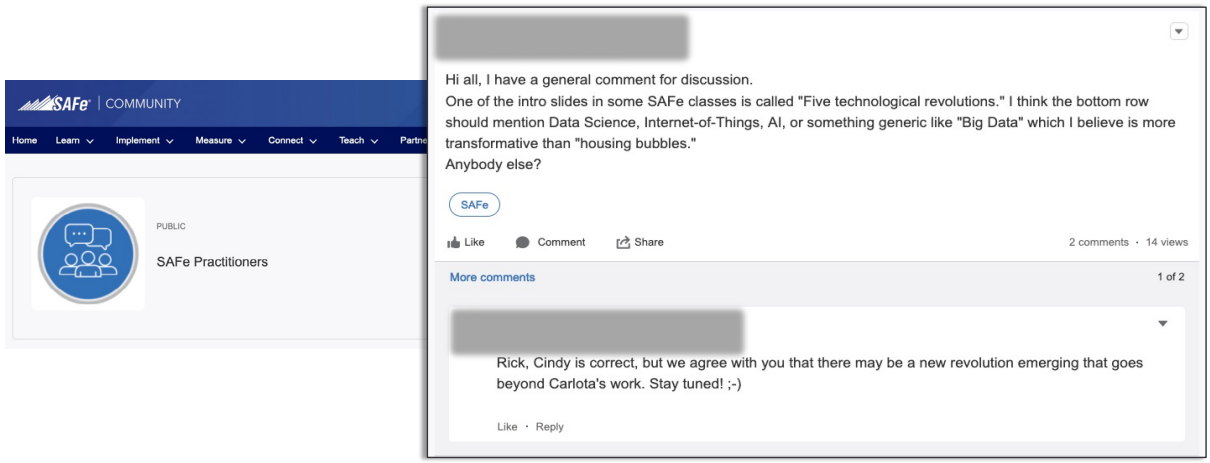
Competency-based self-assessments

Each core competency has a downloadable assessment, along with three Growth recommendations, available in SAFe Collaborate. To access these assets directly, visit the SAFe Community Platform Measure and Grow page: https://bit.ly/Community-MeasureAndGrow

- Organizational Agility
- Lean Portfolio Management
- Enterprise Solution Delivery
- Agile Product Delivery
- **Team and Technical Agility**
- Continuous Learning Culture
- Lean-Agile Leadership

SAFe Forums

Join the SAFe Practitioners Community Forum to connect with a community of SAFe Practitioners



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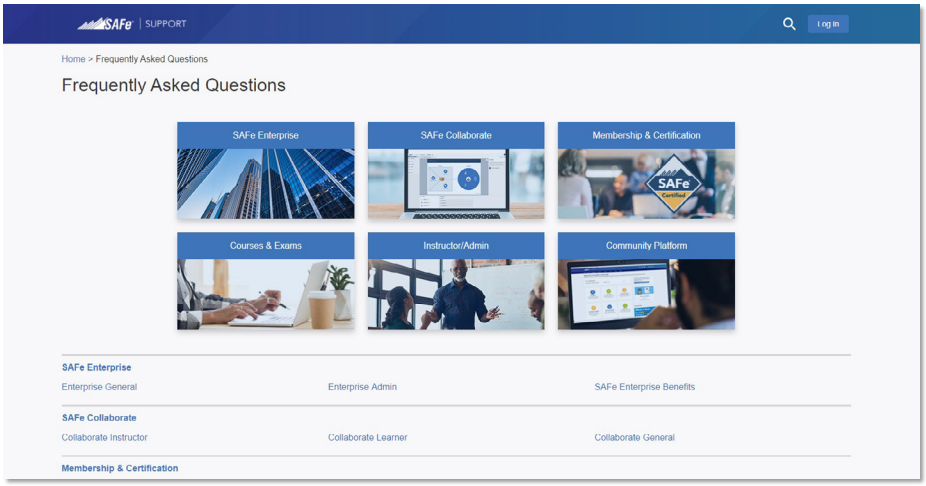
SAFe FAQs

When you need support, check the FAQ page for your question or contact SAI support directly.



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6-13

Good luck on your SAFe Practice with the

SAFe Community Platform!

https://community.scaledagile.com/

SAFe® COMMUNITY

6-14

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SAFe Glossary

SAFe Glossary: Visit the Scaled Agile Framework site (*www.scaledagileframework.com/glossary/*) to download glossaries translated into other languages.

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